

SIZE CHANGE TERMINATION CONDITION FOR INJECTIVE RELATIONS

STEFANO BERARDI

Università di Torino, Torino, Italy
e-mail address: stefano@unito.it

1. INTRODUCTION

We want to study Neil Jones Size Change Termination condition (from now on, SCT) for program termination in a particular case: the case of Injective bi-colored Relations.

Our motivation is that this particular case is enough to treat a relevant application of SCT, namely checking the correctness of a cyclic proof.

We suspect that the algorithm checking SCT is PSPACE-complete also in this case. However we will argue that checking SCT is simpler and faster for injective relations.

Another motivation we have in mind is an automatic translation from cyclic proofs into more ordinary inductive proofs. We prove termination for SCT in the case of injective relations only using first order induction. With respect to a previous attempt, we claim that we can translate cyclic proofs avoiding in the resulting proof a formal proof of Ramsey Theorem and using Intuitionistic Logic only.

2. SIZE CHANGE TERMINATION IN THE CASE OF INJECTIVE RELATIONS

Suppose we have a finite set $X = \{x_1, \dots, x_n\}$ and a finite set $\mathcal{R} = \{(\mathbf{S}_i, \mathbf{P}_i) | i \in [1, r]\}$ of pairs of finite relations on X . We consider the elements of X as variables to which we associate a value in each instant of time.

We will use each relation $a\mathbf{S}_i b$ to express the fact that a is associated in an instant t to a value *greater than or equal* to the value associated to b in the next instant $t + 1$. We will use each relation $a\mathbf{P}_i b$ to express the fact that a is associated in an instant t to a value *greater than* the value associated to b in the next instant $t + 1$. We consider the tree of all finite sequences over $\mathcal{R}$. We call trace any sequence on X and path any sequence on $\mathcal{R}^+$.

We take the words “trace” and “path” from cyclic proofs. Neil Jones uses for the same concept the words “thread” and “multipath”, and states GTC, the Global Trace Condition, as: “all multipaths are in $DESC^\omega$ ”, where $DESC^\omega$ is the set of multipaths having some infinite descending thread.

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Neil Jones terminology is probably more suitable when speaking of programs, while trace and path appear more suitable when speaking of a proof-graph, as we do in this paper. Definitions are the same.

We call value flow any assignment of values in $\mathbf{N}$ to each $x \in X$, the assignment depends on a time parameter t . We require that the set of instants of time is some initial segment of the set $\mathbf{N}$ of natural numbers.

We repeat the same definitions more formally:

Definition 2.1 (Path, Trace and Value Flow). Assume $D = [0, h]$ for some $h \in \mathbf{N}$ or $D = \mathbf{N}$ is any non-empty initial segment of the order on natural numbers.

1. We call any map $b : D \rightarrow \mathcal{R}^+$ a *multipath*, a *path* for short: a path is any finite or infinite sequence over $\mathcal{R}^+$.
2. We call any map $\pi : D \rightarrow X$ a *trace* (or a *thread*): a trace is any finite or infinite sequence over X .
3. We call any map $\theta : D, X \rightarrow \mathbf{N}$ a *value flow*. We call $\theta(t, x) \in \mathbf{N}$ the value of $x \in X$ at time $t \in D$.

We say that a path, trace, value flow is finite when $D = [0, h]$ for some $h \in \mathbf{N}$, that it is infinite when $D = \mathbf{N}$.

We are interested to a condition on $\mathcal{R}$ called the “Global Trace Condition” (GTC), equivalent to the termination of value flows “compatible” with a path of $\mathcal{R}$. GTC is expressed in term of traces.

Definition 2.2 (Global Trace Condition (GTC)). Suppose $X = \{x_1, \dots, x_n\}$, $\mathcal{R} = \{(\mathbf{S}_i, \mathbf{P}_i) | i \in [1, r]\}$ and b, π are a path and a trace of domain D as above.

1. We say that the trace $\pi : D \rightarrow X$ is compatible with the path $b : D \rightarrow \mathcal{R}^+$ on the same D , and we write $\pi \models b$, if and only if : whenever $t, t+1 \in D$ and $b(t) = (\mathbf{S}_i, \mathbf{P}_i) \in \mathcal{R}$ the two consecutive values of the trace are related by $\mathcal{R}$, namely $\pi(t)(\mathbf{S}_i \cup \mathbf{P}_i)\pi(t+1)$. If furthermore we have $\pi(t)\mathbf{P}_i\pi(t+1)$, we say that $t \in D$ is a *b-progress point* of the trace π .
2. $\mathcal{R}$ satisfies the *Global Trace Condition* (GTC) if and only if for all infinite pathes $b : \mathbf{N} \rightarrow \mathcal{R}$ there is some trace $\pi : \mathbf{N} \rightarrow X$ such that $\pi \models b$ and π has infinitely many *b-progress points*.
3. We write $|(\mathbf{S}, \mathbf{P})| = \mathbf{S} \cup \mathbf{P}$ for the *union* of the pair $(\mathbf{S}, \mathbf{P}) \in \mathcal{R}^+$.
4. $\mathcal{R}$ satisfies *injectivity* if and only if for all $i \in [1, r]$ the relation $\mathbf{S}_i \cup \mathbf{P}_i$ is injective.

If formulated as above, the Global Trace condition (GTC) is equivalent to Size Change Termination (SCT), introduced by Neil Jones. SCT was originally used as a sufficient, and quite broad too, condition for termination of recursive calls. When we use GTC as a correctness criterion for cyclic proofs, we *add one more condition*, injectivity of the trace relations, a condition which is missing in the SCT paper.

Injectivity implies that in cyclic proofs two traces of different variables never cross each other and significantly simplifies traces. Instead, in the SCT condition two traces, say, $x_1 \geq x_2 > x_3$ and $y_1 > y_2$, from two different variables x_1, y_1 , can meet into a single variable z . For instance, we can have:

$$x_1 \geq x_2 > x_3 \geq z \quad \text{and} \quad y_1 > y_2 \geq z$$

In this paper we study GTC in the case $\mathcal{R}$ is injective, and from a purely combinatorial viewpoint, as a termination condition. Our result can be used for checking termination of recursive calls, with a faster algorithm but under less general conditions than in Neil

Jones. Our result could be used for translating cyclic proofs into ordinary proofs. Our goal is finding a termination proof using Intuitionistic Logic and First Order Arithmetic.

Several decidable equivalent are known for GTC. In Classical Logic, GTC for $\mathcal{R}$ is equivalent to the fact that all value flows compatible with some path are finite.

We assume injectivity and we are looking for an equivalent version of GTC which can be expressed in First Order Arithmetic, and such that the finiteness of all compatible value flows can be proved in Intuitionistic Arithmetic.

We anticipate that when $n = \text{Card}(X) \geq 1$, we will use lexicographic induction on a length $2n$ sequence $(m_1, r_1, \dots, m_n, r_n)$. For all $k \in [1, n]$, we will have $m_k \in \mathbf{N} \cup \{\infty\}$ and $r_k \in \mathbf{N}$, $r_k < \binom{n}{k}$.

2.1. Comparing our Terminology with the Terminology in Neil Jones SCT Paper.

The SCT paper uses 3-coloring on the set $[1, n] \times [1, n]$. The choice of a color for a pair (i, j) represents what we know about the relation between the value of variable x_i at time t , and the value of variable x_j at time $t + 1$. What we know about the relation can be: either $\geq$ or $>$ or “no information at all”.

We could interpret our relations pairs $(S, P) \in \mathcal{R}$ as the 3-coloring $(S \setminus P, P, X^2 \setminus (S \cup P))$ in the sense of Neil Jones. xSy for $x, y \in X$ represents “the value of x is $\geq$ than the value of y ”, provided we do not have xPy . xPy represents “the value of x is $>$ than the value of y ”. If we do not have xSy nor xPy we represent the fact we do not know what is the relation between the values of x and y .

In order to formulate SCT in our setting, we have first to define the composition of 2-colored relations

2.2. Definition of Composition of 2-colored relations. We define composition in such a way that if $(x, y), (y, z)$ are two pairs, then (x, z) is stationary in the composition relation $(S, P)(S', P')$ if and only if both (x, y) is stationary in S and (y, z) is stationary in S' for some $y \in X$. Dually, (x, z) is progressing in the composition relation $(S, P)(S', P')$ if and only if either (x, y) is progressing in P or (y, z) is progressing in P' or both are, for some $y \in X$.

Definition 2.3 (States, Compositions and Transition Sequences). Assume $X = \{1, \dots, n\}$ and (S, P) is a pair of finite binary relations on X .

1. *Composition on relation pairs.* If $(S, P), (S', P')$ are relation pairs on X we define their composition as

$$(S, P)(S', P') = (SS', SP' \cup PS' \cup PP')$$

2. *(Closure on composition)* Assume $\mathcal{R} \subseteq \mathcal{P}(X \times X)^2$ is any finite set of pairs of finite binary relations on X . $\mathcal{R}^+$ is the set of finite non-empty compositions of elements of $\mathcal{R}$. $\mathcal{R}$ is *composition-closed* if and only if $\mathcal{R} = \mathcal{R}^+$

$\mathcal{R}^+$ is finite and we can compute it, because $n = \text{Card}(X)$ is finite. However, the cardinality of $\mathcal{R}^+$ could be exponential in n . Remark that we defined composition of relation pairs in such a way that it is compatible with the union operator $|\cdot|$ for relation pairs. Indeed we have

$$|(S, P)(S', P')| = |(SS', SP' \cup PS' \cup PP')| = SS' \cup SP' \cup PS' \cup PP' = (S \cup P)(S' \cup P') = |(S, P)| |(S', P')|$$

We can now define SCT.

2.3. The Size Change Termination condition (SCT) and the Progressing Domain Condition (PDC). The SCT condition says: “all bi-colored relations in $\mathcal{R}^+$ have some finite power in which some element progresses to itself”. We write SCT as follows:

$$\forall (\mathbf{S}, \mathbf{P}) \in \mathcal{R}^+. \exists n \in \mathbf{N} \exists x \in X. (x \text{ progresses to } x \text{ in } (\mathbf{S}, \mathbf{P})^n)$$

In previous work we proved that SCT and GTC are equivalent, using Ramsey Theorem.

In this paper we are looking for a more constructive and elementary proof of an equivalent result, namely the termination of all value flows compatible with some path of $\mathcal{R}^+$. The proof should be suitable to a formalization in a proof assistant. We add the injectivity assumption because 2-colored relations with injectivity are enough to describe cyclic proofs, and if we have injectivity proving a termination result requires a weaker hypothesis PDC (see below). Besides, using PDC it is faster checking whether a cyclic proof is correct.

The Progressing Domain Condition (PDC) says: “all bi-colored relations in $\mathcal{R}^+$ have a progressing pair with both elements included in the domain of the relation”, namely

$$\forall (\mathbf{S}, \mathbf{P}) \in \mathcal{R}^+. \exists x, y \in \text{dom}(\mathbf{S} \cup \mathbf{P}). (xPy)$$

PDC is weaker and it is easier to check than the SCT condition. To check PDC, we do not have to look for some x in the progressing to itself in some power of $(\mathbf{S}, \mathbf{P})$. Instead it is enough to find some x progressing to some y , such that both x, y are in the domain of $\mathbf{S} \cup \mathbf{P}$.

However, PDC has the drawback that it implies GTC and termination only for *injective* bi-colored relations. We explain why.

2.3.1. PDC and composition-closure do not imply GTC, in the case of non injective family of bi-colored relations. Here is a counter-example $\mathcal{R}_1$. Let $X = \{x_1, x_2\}$, $\mathcal{R}_1 = \{(\mathbf{S}, \mathbf{P})\}$ and $\mathbf{S} = \{(x_1, x_1)\}$, $\mathbf{P} = \{(x_2, x_1)\}$. Then $\mathcal{R}_1$ is composition-closed because $(\mathbf{S}, \mathbf{P})(\mathbf{S}, \mathbf{P}) = (\mathbf{SS}, \mathbf{SP} \cup \mathbf{PP} \cup \mathbf{PS}) = (\text{by } \mathbf{SP} = \mathbf{PP} = \emptyset)(\mathbf{SS}, \mathbf{PS}) = (\mathbf{S}, \mathbf{P})$. The unique coloring $(\mathbf{S}, \mathbf{P})$ satisfies PDC because $x_2 \in \text{dom}((\mathbf{S}, \mathbf{P}))$ is progressing to $x_1 \in \text{dom}((\mathbf{S}, \mathbf{P}))$. However, $\mathcal{R}_1$ does not satisfy GTC, because the unique infinite path is $\pi = ((\mathbf{S}, \mathbf{P}), (\mathbf{S}, \mathbf{P}), (\mathbf{S}, \mathbf{P}), \dots)$, and any infinite trace compatible with π is either $x_1x_1x_1\dots$ or $x_2x_1x_1x_1\dots$. Thus, any infinite trace compatible with π progresses *at most once*. The implication $\text{PDC} \implies \text{GTC}$ is false because $\mathcal{R}_1$ is not injective. Indeed, the relation $R = |(\mathbf{S}, \mathbf{P})| = \{(x_1, x_1), (x_2, x_1)\}$ is not injective.

Instead, we will prove that $\text{PDC} + \text{injectivity}$ imply finiteness of value flows, or, using Classical Logic, GTC:

Theorem 2.4 (Finiteness of $\mathcal{R}$ -compatible transitions 1). If $\mathcal{R}$ is a finite list of bi-colored *injective* relations satisfying Progressing Domain Condition (PDC), then any value flow compatible with $\mathcal{R}$ is finite.

We postpone the proof to **S 2.5**. Before we remark that the Termination Theorem for $\mathcal{R}$ can be reformulated as an induction schema on $\mathbf{N}^n$, as follows.

Let $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$. For any $(\mathbf{S}, \mathbf{P}) \in \mathcal{R}^+$ we define a relation $<_{(\mathbf{S}, \mathbf{P})}$. We define $\vec{x} <_{(\mathbf{S}, \mathbf{P})} \vec{y}$ if and only if:

$$\forall i, j \in [1, n]. (i\mathbf{S}j \implies x_i \leq y_j) \wedge (i\mathbf{P}j \implies x_i < y_j)$$

We define $<_{\mathcal{R}}$ as the union $\bigcup \{<_{(\mathbf{S}, \mathbf{P})} \mid (\mathbf{S}, \mathbf{P}) \in \mathcal{R}^+\}$ of all relations $<_{(\mathbf{S}, \mathbf{P})}$ for $(\mathbf{S}, \mathbf{P}) \in \mathcal{R}^+$. Then Thm. 2.4 can be reformulated as follows. Assume PDC and injectivity for $\mathbf{S} \cup \mathbf{P}$, for all $(\mathbf{S}, \mathbf{P}) \in \mathcal{R}^+$. Then we can prove **the induction schema for $<_{\mathcal{R}}$** .

More in detail, using Progressing Domain Condition we will reformulate a cyclic proof as an ordinary proof by adding first order intuitionistic reasoning and induction on $[0, \infty]$

nested n times, where $n = \text{Card}(X)$ is the cardinality of the set of variables, together with induction on the finite sets $[1, \binom{n}{k}[$ nested n times, for $k = 1, \dots, n$.

A last remark: in the definition of PDC we cannot simplify $x, y \in \text{dom}(\mathbf{S} \cup \mathbf{P})$ to the weaker condition $x, y \in X$. Let us call “Progressing Condition” or PC the resulting condition. Then PC is weaker than PDC, and indeed PC just says: $\mathbf{P} \neq \emptyset$ for all $(\mathbf{S}, \mathbf{P}) \in \mathcal{R}^+$. If we have PC in the place of PDC then the Termination Theorem fails, even if we assume composition-closure and injectivity. We explain why.

2.3.2. PC, composition-closure and injectivity does not imply GTC. Here is a counter-example $\mathcal{R}_2$. Let $X = \{x_1, x_2\}$, $\mathcal{R}_2 = \{(\mathbf{S}, \mathbf{P})\}$ and $\mathbf{S} = \{(x_1, x_1)\}$, $\mathbf{P} = \{(x_1, x_2)\}$. Then $\mathcal{R}_2$ is composition-closed because $(\mathbf{S}, \mathbf{P})(\mathbf{S}, \mathbf{P}) = (\mathbf{SS}, \mathbf{SP} \cup \mathbf{PP} \cup \mathbf{PS}) = (\text{by } \mathbf{PP} = \mathbf{PS} = \emptyset)(\mathbf{SS}, \mathbf{SP}) = (\mathbf{S}, \mathbf{P})$. $\mathcal{R}_2$ is injective, because the relation $R = |(\mathbf{S}, \mathbf{P})| = \{(x_1, x_1), (x_1, x_2)\}$ is injective. The unique coloring $(\mathbf{S}, \mathbf{P})$ satisfies PC because $x_1 \in \text{dom}((\mathbf{S}, \mathbf{P}))$ is progressing to x_2 . Remark that $x_2 \notin \text{dom}((\mathbf{S}, \mathbf{P}))$ and x_1 does not progress to x_1 , therefore PDC fails while PC holds. $\mathcal{R}_2$ does not satisfy GTC, because the unique infinite path is $\pi = (R, R, R, R, \dots)$, and the unique infinite trace in π is $x_1 x_1 x_1 \dots$, therefore the unique infinite trace never progresses.

In order to prove the Termination Theorem we have to define States and Transition chains first.

2.4. Definition of States and Transitions chain. We define the set $\mathcal{S}$ of computation states and the set $\mathcal{T}$ of transition chains from a computation state to another one.

Definition 2.5 (States, Compositions and Transition Sequences). Assume $X = \{x_1, \dots, x_n\}$ and $(\mathbf{S}, \mathbf{P})$ is a pair of finite binary relations on X .

1. We denote with $\mathcal{V}$ the set of *value assignment* $\sigma : X \rightarrow \mathbf{N}$ for variables.
2. We denote with $\mathcal{S} = \mathcal{R}^+ \times \mathcal{V}$ the set of *computation states*, each $s = ((\mathbf{S}, \mathbf{P}), \sigma)$ consists of a pair $(\mathbf{S}, \mathbf{P})$ of finite binary relations on X and a value assignment $\sigma : X \rightarrow \mathbf{N}$.
3. (*Transition*) Assume $s = ((\mathbf{S}, \mathbf{P}), \sigma)$, $s' = ((\mathbf{S}', \mathbf{P}'), \sigma')$, $s, s' \in \mathcal{S}$ are states. We call (s, s') a *transition* if and only if $\forall(x, y) \in \mathbf{S}. \sigma(x) \geq \sigma'(y)$ and $\forall(x, y) \in \mathbf{P}. \sigma(x) > \sigma'(y)$.
4. (*Transition Sequence*) A *transition sequence* is any sequence $(s_1, \dots, s_n) \in \mathcal{S}$ of states such that (s_i, s_{i+1}) is a transition for all $i \in [1, n[$. We denote the set of transition sequences with $\mathcal{T}$.

2.5. A Proof of the Termination Theorem 2.4. Assume that $\mathcal{R}$ is a composition-closed set, that $\mathcal{R}$ is injective and that $\mathcal{R}$ satisfies PDC, namely: for all $(\mathbf{S}, \mathbf{P}) \in \mathcal{R}$ we have $x\mathbf{P}y$ for some $x, y \in \text{dom}(\mathbf{S} \cup \mathbf{P})$. Let $\mathcal{T}$ be the set of transition sequences for $\mathcal{R}$. We want to prove the Termination Theorem: all transition sequences terminate.

We will prove using Intuitionistic Logic and First Order Arithmetic that the tree $\mathcal{T}$ of transition sequences (having root the empty sequence) satisfies the induction schema.

To this aim, let $[0, \infty] = \mathbf{N} + \{\infty\}$ be $\mathbf{N}$ extended with an “infinite” point ∞ , and with the relation $n < \infty$ for all $n \in \mathbf{N}$. Let

$$\mathcal{L} = \prod_{k=1}^{k=n} [0, \infty] \times [0, \binom{n}{k}[$$

with the lexicographic ordering. We define a map $f : \mathcal{T} \rightarrow \mathcal{L}$ and we prove that f is increasing w.r.t. the one-step extension $>_1$ in $\mathcal{T}$ and the lexicographic ordering $<_{\text{lex}}$ on $\mathcal{L}$.

Namely we will prove that if $\theta \in \mathcal{T}$ and $\eta \in \mathcal{T}$, $\eta >_1 \theta$ then $f(\eta) <_{\text{lex}} f(\theta)$. This implies that all transition sequences in $\mathcal{T}$ terminate. We use the following folk-lore Proposition.

We need some more ingredients for a proof of the Termination Theorem 2.4. The first one is some sufficient condition for an injective relation R on X to be a bijection.

Proposition 2.6 (Injective Relations and Cardinality). Assume that $R \subseteq X \times Y$ is everywhere defined and is injective. Then

1. $\text{Card}(X) \leq \text{Card}(Y)$
2. if $\text{Card}(X) = \text{Card}(Y)$ then R is a bijection between X and Y .

Proof. Assume $R \subseteq X \times Y$ is everywhere defined and injective.

1. Then for all $x \in X$ the sets $R(x) = \{y \in Y \mid R(x, y)\}$ are inhabited, because R is everywhere defined, and are pairwise disjoint, because R is injective. We deduce that $\text{Card}(X) \leq \text{Card}(Y)$.
2. Now suppose that $\text{Card}(X) = \text{Card}(Y)$. Then all sets $R(x)$ have cardinality 1, otherwise we would have $\text{Card}(X) < \text{Card}(Y)$. Thus, R is a function, and R is injective by assumption. Now we use the well-known fact that any injection between two *finite* sets of equal cardinality is a bijection.

□

We also need a sufficient condition for a relation R being a bijection between $\text{dom}(RS)$ and $\text{dom}(S)$: R should be injective and the cardinalities of $\text{dom}(RS)$ and $\text{dom}(S)$ should be the same. We also provide a sufficient condition for R being a bijection between $\text{dom}(R)$ and $\text{dom}(RS)$: R should be injective and $\text{dom}(R) = \text{dom}(RS) = \text{dom}(S)$.

For any binary relation R on X we write $R[Y, Z]$ for $R \cap (Y \times Z)$, the *restriction of R to Y, Z* and $R[Y]$ for $R \cap (Y \times Y)$, the *restriction of R to Y* .

Lemma 2.7 (Composition of Injective Relations and Cardinality). Assume that R, S are finite binary relations on $X = \{x_1, \dots, x_n\}$.

1. $\text{dom}(R) \supseteq \text{dom}(RS)$.
2. If R is injective then $R \cap (\text{dom}(RS) \times \text{dom}(S))$ is everywhere defined and surjective, and $\text{Card}(\text{dom}(RS)) \leq \text{Card}(\text{dom}(S))$
3. If R is injective and $\text{Card}(\text{dom}(RS)) = \text{Card}(\text{dom}(S))$ then $R \cap (\text{dom}(RS) \times \text{dom}(S))$ is a bijection.
4. If R is injective and $\text{dom}(R) = \text{dom}(RS) = \text{dom}(S)$ then $R[\text{dom}(R)]$ is a bijection.

Proof. 1. We have to prove that $\text{dom}(R) \supseteq \text{dom}(RS)$. We assume $x \in \text{dom}(RS)$ in order to prove that $x \in \text{dom}(R)$. If $x \in \text{dom}(RS)$ then by definition of RS we have $(x, y) \in \text{dom}(R)$ and $(y, z) \in \text{dom}(S)$ for some $y, z \in X$. We conclude that $x \in \text{dom}(R)$, as wished.

2. By Lemma 2.6. 1. if R is injective then in order to prove that $\text{Card}(\text{dom}(RS)) \leq \text{Card}(\text{dom}(S))$ it is enough to prove two conditions on $R \cap (\text{dom}(RS) \times \text{dom}(S))$:

- (1) $R \cap (\text{dom}(RS) \times \text{dom}(S))$ is everywhere defined. For, we assume that $x \in \text{dom}(RS)$ in order to prove that for some $(x, y) \in R$ we have $y \in \text{dom}(S)$. From $x \in \text{dom}(RS)$ we deduce that there is some $z \in X$ such that $(x, z) \in RS$, therefore for some $y \in X$ we have $(x, y) \in R$ and $(y, z) \in S$. We conclude that $(x, y) \in R$ and $y \in \text{dom}(S)$, as we wished to prove.

- (2) $R \cap (\text{dom}(RS) \times \text{dom}(S))$ is injective. This is because this relation is a subset of R which is injective.
3. By point 2. above and Proposition 2.6.2..
4. Assume $\text{dom}(R) = \text{dom}(RS)$ and $\text{dom}(R) = \text{dom}(S)$ and R injective in order to prove that $R \upharpoonright \text{dom}(R)$ is a bijection. $\text{dom}(R) = \text{dom}(S)$ implies that $\text{Card}(\text{dom}(R)) = \text{Card}(\text{dom}(S))$. By $\text{Card}(\text{dom}(R)) = \text{Card}(\text{dom}(S))$, R injective and point 3. above we deduce that $R \upharpoonright \text{dom}(RS), \text{dom}(S)$ is a bijection. The thesis now follows from $\text{dom}(RS) = \text{dom}(R)$ and $\text{dom}(S) = \text{dom}(R)$. $\square$

Remark that the fact that $R \upharpoonright \text{dom}(R)$ is a bijection does not mean that R itself is a bijection. For instance, an element $x \in \text{dom}(R)$ can have some image $y \notin \text{dom}(R)$. In this case x has at least two images and R is no bijection, while $R \upharpoonright \text{dom}(R)$ is.

We define the value $\text{val}(Y, \sigma)$ of a subset $Y \subseteq X$ of X w.r.t. a valuation $\sigma : X \rightarrow \mathbb{N}$ as the sum of all values of σ on X

$$\text{val}(Y, \sigma) = \sum_{y \in Y} \sigma(y)$$

Lemma 2.8 (The value of progressing bijection is decreasing). Assume that $r = (\mathbf{S}, \mathbf{P})$ is a relation pair of $X = \{x_1, \dots, x_n\}$ and $R = |r| = \mathbf{S} \cup \mathbf{P}$. Assume $\sigma, \tau \in \mathcal{V}$ are value assignments on X and $Y, Z \subseteq X$. Assume that:

1. $R \cap (Z \times Y)$ is a bijection
2. we have “weak progression of σ to τ ”: for all $i \in Y, j \in Z. ((i, j) \in R \implies \sigma(i) \geq \tau(j))$.
3. we have “strict progression of σ to τ ”: for some $i \in Y, j \in Z$ we have $(i, j) \in R$ and $\sigma(i) > \tau(j)$.

Then $\text{val}(Y, \sigma) > \text{val}(Z, \tau)$.

Proof. From $R \cap (Z \times Y)$ bijection we deduce that $\text{Card}(Y) = \text{Card}(Z) = p$ for some $p \in [0, n]$. We also deduce that $Y = \{i_1, \dots, i_p\}$ and $Z = \{j_1, \dots, j_p\}$ for some increasing sequences $i_1 < \dots < i_p$ and $j_1 < \dots < j_p$ on $[1, n]$ such that $(i_h, j_h) \in R$ for all $h \in [1, p]$.

By weak and strict progression of σ to τ we deduce that $\sigma(i_h) \geq \tau(j_h)$ for all $h \in [1, p]$ and $\sigma(i_h) > \tau(j_h)$ for some $h \in [1, p]$. We conclude that $\text{val}(Y, \sigma) = \sum_{h \in [1, p]} \sigma(i_h) > \sum_{h \in [1, p]} \tau(j_h) = \text{val}(Z, \tau)$, as wished. $\square$

3. TRANSLATING GLOBAL TRACE CONDITION INTO NESTED INDUCTION (FOR INJECTIVE RELATIONS AND ANY CARDINALITY)

Suppose $\text{Card}(X) = n$. Suppose $\mathcal{R} = \{(\mathbf{S}_i, \mathbf{P}_i) | i \in [1, a]\}$ is any finite set of bi-colorings on finite relations on X , closed by bi-color composition, i.e., $\mathcal{R} = \mathcal{R}^+$. Suppose $\mathcal{R}$ satisfies PDC, the progressing cycle condition.

Assume $\pi : [1, n] \rightarrow \mathcal{R}$ is any length n list on $\mathcal{R}$, with each $\pi(j) = (\mathbf{S}_{i_j}, \mathbf{P}_{i_j})$ some element of $\mathcal{R}$. Let Π be the set of such lists on $\mathcal{R}$.

Let $\theta : [1, n+1] \rightarrow (X \rightarrow \mathbb{N})$ any family of assignment compatible with π : we require that $\theta(i) \pi(i)$ -progresses to $\theta(i+1)$, for all $i \in [1, n]$. To put otherwise we have

$$\theta(1) (\mathbf{S}_{i_1}, \mathbf{P}_{i_1}) \theta(2) (\mathbf{S}_{i_2}, \mathbf{P}_{i_2}) \theta(3) \dots \theta(n) (\mathbf{S}_{i_n}, \mathbf{P}_{i_n}) \theta(n+1)$$

Let Θ be the set of compatible pairs (π, θ) . We define the component-wise one-step extension on Θ by: $(\pi, \theta) <_1 (\pi', \theta')$ if and only if $\pi <_1 \pi'$ and $\theta <_1 \theta'$ if and only if $\exists S, P \in \mathcal{R}_{\text{fin}}(X^2)(\pi' = \pi \star (S, P))$ and $\exists \sigma : X \rightarrow \mathbf{N}(\theta' = \theta \star \sigma)$.

We associate a weight $w(\pi, \theta) \in \Pi_{k=1}^{k=n}(\mathbf{N} \cup \{\infty\} \times [1, \binom{n}{k}])$ in such a way that if $(\pi, \theta) <_1 (\pi', \theta')$ then $w(\pi, \theta) > w(\pi', \theta')$ in the lexicographic ordering.

Assume $L : [1, l] \rightarrow \mathcal{R}_{\text{fin}}(\mathbf{N} \times \mathbf{N})$ is a finite relation sequence, with $R_i = L(i)$ for all $i \in [1, l]$. Then for all $i, j \in [1, l]$, $i \leq j$ we define $R_{i,j} = \Pi_{h=i}^{h=j} R_h = R_i \dots R_j$.

By Lemma 1. and $\text{dom}(R_{i,j+1}) = \text{dom}(R_{i,j} R_{j+1})$ we have $\text{dom}(R_{i,j}) \supseteq \text{dom}(R_{i,j+1})$, that is, the sequence of sets $\text{dom}(R_{i,j})$ for $j \in [i, l]$ is weakly decreasing. When all these sequences are constant we say that the list L is domain-stable. Here is the formal definition.

Definition 3.1 (Domain-Stability for a Sequence). 1. A finite relation sequence $L : [1, l] \rightarrow \mathcal{R}_{\text{fin}}(X \times X)$ is domain-stable if and only if all $i \in [1, l]$ we have $\text{Card}(\text{dom}(R_i)) = \text{Card}(\text{dom}(R_{i,l}))$.
2. A finite sequence $\pi : [1, l] \rightarrow \mathcal{R}_{\text{fin}}(X \times X)^2$ of relation pairs is *domain-stable* if the associated relation sequence $\{\pi(i) \mid i \in [1, l]\}$, obtained by taking the union of each pair, is domain-stable.

If $L = \{R_i \mid i \in [1, l]\}$ is domain-stable and $i \in [1, l]$, then the first set $\text{dom}(R_{i,i})$ and the last set $\text{dom}(R_{i,l})$ in the weakly decreasing sequence $\{\text{dom}(R_{i,j}) \mid j \in [i, l]\}$ are equal. Thus, if L is domain stable then all these sequences are constant. The next Lemma describes in details the structure of a domain-stable sequence π on a set $\mathcal{R}$ of relation pairs.

Lemma 3.2 (Domain-stable injective bi-coloring and the Progressing Domain condition). Assume that $\pi : [1, l] \rightarrow \mathcal{R}_{\text{fin}}(X \times X)^2$ is a *domain-stable* finite list of *injective* pairs of relations on X , compatible with the assignment list $\theta : [1, l] \rightarrow (X \rightarrow \mathbf{N})$.

For all $i \in [1, l]$ suppose that $R_i = |\pi(i)|$. For all $j \in [1, l]$, $i \leq j$ suppose that $R_{i,j} = \Pi_{h=i}^{h=j} R_h = R_i \dots R_j$.

1. If $i, j \in [1, l]$, $i < j$ we have $\text{Card}(\text{dom}(R_i)) \leq \text{Card}(\text{dom}(R_j))$. If furthermore $\text{Card}(\text{dom}(R_i)) = \text{Card}(\text{dom}(R_j))$, then $R_{i,j-1} \upharpoonright \text{dom}(R_i), \text{dom}(R_j)$ is a bijection from $\text{dom}(R_i)$ to $\text{dom}(R_j)$.
2. If $i, j \in [1, l]$, $i < j$ and $\text{Card}(\text{dom}(R_i)) = \text{Card}(\text{dom}(R_j))$, then $R_{i,j-1} \upharpoonright \text{dom}(R_i), \text{dom}(R_j)$ is a bijection, the value of $\theta(i)$ on any $x \in \text{dom}(R_i)$ is $\geq$ than the value of $\theta(j)$ on the corresponding point $y \in \text{dom}(R_j)$ and $\text{val}(\text{dom}(R_i), \theta(i)) \geq \text{val}(\text{dom}(R_j), \theta(j))$.
3. If $i, j \in [1, l]$, $i < j$ and $\text{dom}(R_i) = \text{dom}(R_j)$, and the composition closure of $\{\pi(i) \mid i \in [1, l]\}$ satisfies the Progressing-domain condition, then $\text{val}(\text{dom}(R_i), \theta(i)) > \text{val}(\text{dom}(R_j), \theta(j))$.
4. Let $k \in [1, n]$ with $n = \text{Card}(X)$. Assume that the composition closure of $\mathcal{R} = \{\pi(i) \mid i \in [1, l]\}$ satisfies the Progressing-domain condition. Then the number of indexes $i \in [1, l]$ such that $\text{Card}(\text{dom}(R_i)) = k$ and $\text{val}(\text{dom}(R_i), \theta(i)) = v$ is $\leq \binom{n}{k}$, for all $v \in \mathbf{N}$.

Proof. 1. If L is a domain-stable sequence of injective relations then by Lemma 2.7. 2., 3. we deduce that $\text{Card}(\text{dom}(R_i R_{i+1})) \leq \text{Card}(\text{dom}(R_{i+1}))$ and if $\text{Card}(\text{dom}(R_i R_{i+1})) = \text{Card}(\text{dom}(R_{i+1}))$ that $R_i \upharpoonright \text{dom}(R_i R_{i+1}), \text{dom}(R_{i+1})$ is a bijection.

For all $i \in [1, l-1]$ by stability we have $\text{dom}(R_i) = \text{dom}(R_i R_{i+1})$, therefore $\text{Card}(\text{dom}(R_i)) \leq \text{Card}(\text{dom}(R_{i+1}))$ and if $\text{Card}(\text{dom}(R_i)) = \text{Card}(\text{dom}(R_{i+1}))$ then $R_i \upharpoonright \text{dom}(R_i), \text{dom}(R_{i+1})$ is a bijection.

2. By the previous point we know that $R_{i,j-1} \upharpoonright \text{dom}(R_{i,j-1})$ is a bijection between $\text{dom}(R_i)$ and $\text{dom}(R_j)$. By compatibility between π and θ , the value of $\theta(i)$ on any $x \in \text{dom}(R_i)$ is $\geq$ than the value of $\theta(j)$ on the corresponding point $y \in \text{dom}(R_j)$. Thus, $\text{val}(\text{dom}(R_i), \theta(i)) \geq \text{val}(\text{dom}(R_j), \theta(j))$.
3. Assume $\text{dom}(R_i) = \text{dom}(R_j)$. Then $\text{Card}(\text{dom}(R_i)) = \text{Card}(\text{dom}(R_j))$ and the value of $\theta(i)$ on any $x \in \text{dom}(R_i)$ is $\geq$ than the value of $\theta(j)$ on the corresponding point $y \in \text{dom}(R_j)$. By the Progressing Domain Condition, $R_{i,j-1}$ progresses on $\text{dom}(R_{i,j-1})$, that is, on $\text{dom}(R_i)$, that is, between $\text{dom}(R_i)$ and $\text{dom}(R_i) = \text{dom}(R_j)$. By compatibility between π and θ , the value of $\theta(i)$ on some $x \in \text{dom}(R_i)$ is $>$ than the value of $\theta(j)$ on the corresponding point $y \in \text{dom}(R_j)$. We conclude that $\text{val}(\text{dom}(R_i), \theta(i)) > \text{val}(\text{dom}(R_j), \theta(j))$.
4. Assume $i, j \in [1, l]$, $i < j$ and $\text{Card}(\text{dom}(R_i)) = \text{Card}(\text{dom}(R_j)) = k$ and $\text{val}(\text{dom}(R_i), \theta(i)) = \text{val}(\text{dom}(R_j), \theta(j)) = v$. By the previous point we deduce that $\text{dom}(R_i) \neq \text{dom}(R_j)$.

There are $\binom{n}{k}$ different cardinality k subsets of an n element set X , therefore at most $\binom{n}{k}$ indexes for $i \in [1, l]$ such that $\text{Card}(\text{dom}(R_i)) = k$ and $\text{val}(\text{dom}(R_i), \theta(i)) = v$. □

Proof of Termination Theorem 1, 2.4. We first define a list of n weights $w(\pi, \theta)$ for a length m sequence π with stable domains. We associate to each cardinality $k \in [1, m]$ a single weight.

Suppose there is some $i \in [1, m]$ such that $\text{Card}(\text{dom}(|\pi(i)|)) = k$. Then we define $w(\pi, \theta)_k = (m_k, r_k)$ where m_k is the least value for $\text{val}(\pi(i), \theta(i))$ such that $\text{Card}(\text{dom}(|\pi(i)|)) = k$. We define $r_k = \binom{n}{k} - s_k$, where $s_k \geq 1$ is the number of $i \in [1, m]$ such that $\text{val}(\pi(i), \theta(i)) = m_k$ and $\text{Card}(\text{dom}(|\pi(i)|)) = k$ ¹

If there is no $i \in [1, m]$ such that $\text{Card}(\text{dom}(|\pi(i)|)) = k$ then we set $w(\pi, \theta)_k = (\infty, 0)$.

We define $w(\pi, \theta)$ as the sequence $(w(\pi, \theta)_1, \dots, w(\pi, \theta)_n)$ of weights.

In each domain stable segment by Lemma 2. we have $\text{Card}(\text{dom}(R_{i,m})) = \text{Card}(\text{dom}(R_i R_{i+1,m})) \leq \text{Card}(\text{dom}(R_{i+1,m}))$. By stability we have $\text{dom}(R_i) = \text{dom}(R_{i,m})$ and $\text{dom}(R_{i+1}) = \text{dom}(R_{i+1,m})$, we deduce $\text{Card}(\text{dom}(R_i)) \leq \text{Card}(\text{dom}(R_{i+1}))$.

For all $k \in [1, n]$ and any domain stable $\pi : [1, m] \rightarrow \mathcal{R}$ we define $S(\pi, k) = \{i \in [1, m] \mid \text{Card}(\text{dom}(R_i)) = k\}$. Each We claim that $S(\pi, k)$ is a possibly empty segment. Indeed, if $i, j \in S(\pi, k)$, $i \leq h \leq j$ then $k = \text{Card}(\text{dom}(R_i)) \leq \text{Card}(\text{dom}(R_h)) \leq \text{Card}(\text{dom}(R_j)) = k$, therefore $\text{Card}(\text{dom}(R_h)) = k$ and $h \in S(\pi, k)$. From $\text{Card}(\text{dom}(R_i))$ weakly increasing we deduce that $S(\pi, k)$ is to the left of $S(\pi, k')$ whenever $k, k' \in [1, n]$ and $k < k'$.

We can decompose each domain-stable $q : [1, m] \rightarrow \mathcal{R}$ into a consecutive composition $q_1 * \dots * q_n$, with each q_k the possibly empty restriction of q to $S(\pi, k)$.

Assume $L = (c_1, \dots, c_n)$ is a coloring list. We define a re-grouping of L as any coloring sequence $M = (c_1 \dots c_{a_1}, c_{a_1+1} \dots c_{a_2}, \dots, c_{a_{m-1}+1} \dots c_n)$ of composition of 2-colorings takne

¹ r_k is a bound to the number of times the values $(m_1, r_1, \dots, m_k)$ can repeat equal if we extend π , and $r_k = 0$ means these values cannot repeat. Indeed, we will prove that if we consecutively have $\text{Card}(\text{dom}(|\pi(i)|)) = k$ and $\text{val}(\pi(i), \theta(i)) = m_k$, then the set $Y = \text{dom}(|\pi(i)|)$ cannot repeat. Y is any subset of k elements of $[1, n]$, therefore the number of possible values for Y starts from $\binom{n}{k}$ and decreases by 1 in each step, and it cannot go below zero.

from L , with the same ordering as in L , and whose product is the same as L : $\Pi M = \Pi L$. M is obtained by partitioning L into adjacent segments $[a_1, a_2 - 1], [a_2, a_3 - 1], \dots, [a_{m-1}, a_m - 1], [a_m, n]$, with $a_1 = 1$, then composing the coloring of each segment $[a_i, a_{i+1} - 1]$ into a single coloring. Say: we can group $L = (c_1, c_2, c_3, c_4, c_5, c_6)$ into $M = (c_1 c_2, c_3, c_4, c_5, c_6)$, and in many more ways. We call $a = a_1, \dots, a_m$ the adjacence sequence. If $n \geq 1$ there are 2^{n-1} coloring, exactly the number of subsets $\{a_1, \dots, a_m\}$ of $[1, n]$ including $a_1 = 1$.

Formally, we define re-grouping and the adjacence sequence by induction on n . If $n = 0$ then $M = ()$ is the only re-grouping of $L = ()$ with adjacence sequence $a = ()$. Suppose $n \geq 1$ and we defined re-grouping and adjacence for all lists of length $\leq n - 1$ and $L = L_1 \star L_2 \star (c_n)$ is any decomposition of L into two consecutive sublists plus one element. Then L_1 has length $\leq n - 1$. If M is any re-grouping of L_1 with adjacence sequence a and $d = \Pi L_1 \star (c_n)$ is the composition of all colorings in the list $L_2 \text{stat}(c_n)$ then $M \star (d)$ is one of the re-groupings of L , the re-grouping with adjacence sequence $a \star (\text{len}(L_2) + 1)$

Assume (π, θ) is any compatible pair bi-coloring sequence/assignment sequence with $\pi : [1, n] \rightarrow \mathcal{R}$ and $\theta : [1, n], X \rightarrow \mathbf{N}$. We define a map $c(\pi, \theta) = (q, \eta)$ returning a compatible pair bi-coloring sequence/assignment sequence, and for some increasing map $a : [1, m] \rightarrow [1, n]$ we have:

1. $q : [1, m] \rightarrow \mathcal{R}$ is a re-grouping of π with adjacence sequence a ,
2. $\eta = \theta \circ a$, namely $\eta(t, x) = \theta(a(t), x)$ for all $t \in [1, m], x \in X$.
3. q has stable domain and
4. $(\pi, \theta) <_1 (\pi', \theta')$ implies $w(c(\pi, \theta)) >_{\text{lex}} w(c(\pi', \theta'))$ in the lexicographic ordering.

More in detail we will have $c(\pi, \theta) = (q, \eta)$, $c(\pi, \theta) = (q, \eta)$ $c(\pi', \theta') = (q', \eta')$ and $q = q_1 * \dots * q_n$, $q' = q'_1 * \dots * q'_n$ are decomposition with each q_k, q'_k a sequence of 2-colored relations whose domains all have cardinality k , then for some $k \in [1, n]$ we have $q_1 = q'_1, \dots, q_{k-1} = q'_{k-1}$ and $q_k <_1 q'_k$ and $q' = q'_1 \dots q'_k$. This explains why we have $w(c(\pi, \theta)) >_{\text{lex}} w(c(\pi', \theta'))$.

Suppose $\pi' = \pi \star (\mathbf{S}, \mathbf{P})$ and $\theta' = \theta \star \sigma$ and $c(\pi, \theta) = (q, \eta)$ with $\text{len}(q) = \text{len}(\eta) = m$ in order to define $c(\pi', \theta') = (q', \eta')$, sequences of length m' . We distinguish two cases.

1. Assume that $q \star (\mathbf{S}, \mathbf{P})$ is stable. Then we set $q' = q \star (\mathbf{S}, \mathbf{P})$ and $\eta' = \eta \star (n + 1)$, sequences of length $m' = m + 1$.
2. Assume that $q \star (\mathbf{S}, \mathbf{P})$ is *not* stable. Then we take the first $m' \in [1, m]$ such that $\text{dom}(q(m') \circ \dots \circ q(m)) \subset \text{dom}(q(m') \circ \dots \circ q(m) \circ (\mathbf{S}, \mathbf{P}))$. We define $q'(i) = q(i)$ for all $i \in [1, m' - 1]$ and $q'(m') = q(m') \circ \dots \circ q(m) \circ (\mathbf{S}, \mathbf{P})$. We define $\eta'(i) = \eta(i)$ for all $i \in [1, m']$.

This is $c(\pi', \theta')$.

In any domain-stable sequence of injective relations the cardinality sequence $\text{Card}(\text{dom}(|q(i)|))$ is equal to $\text{Card}(\text{dom}(|q(i)| \dots |q(m)|))$ and therefore it is weakly increasing in the index $i \in [1, m]$.

We prove that $w(c(\pi, \theta)) > w(c(\pi', \theta'))$ by cases on the definition of $c(\pi', \theta')$.

1. Assume that $q \star (\mathbf{S}, \mathbf{P})$ is stable. Then $w(c(\pi, \theta))_l = w(c(\pi', \theta'))_l$ for all $l < m'$. If $\text{dom}(\pi'(m')) = \text{dom}(\pi'(m))$ we use Lemma 3.2. 3. to prove that $w(c(\pi, \theta))_{m'} > w(c(\pi', \theta'))_{m'}$. If $\text{dom}(\pi'(m')) \neq \text{dom}(\pi'(m))$ we use Lemma 3.2. 4. to prove that $w(c(\pi, \theta))_{m'} > w(c(\pi', \theta'))_{m'}$.
2. Assume that $q \star (\mathbf{S}, \mathbf{P})$ is *not* stable. Then $w(c(\pi, \theta))_l = w(c(\pi', \theta'))_l$ for all $l < m'$ and $w(c(\pi, \theta))_{m'} > w(c(\pi', \theta'))_{m'}$ because the domain of $q'(m')$ is smaller than the domain of

$q(m')$. In the case $\text{Card}(\text{dom}(q'(m'))) > \text{Card}(\text{dom}(q(1)), \dots, \text{Card}(\text{dom}(q(m'-1)))$ we have $w(c(\pi, \theta))_{m'} = \infty > w(c(\pi', \theta'))_{m'}$. In the case $\text{Card}(\text{dom}(q'(m'))) = \text{Card}(\text{dom}(q(m'-1)))$, by arguing as in the previous case we prove that q' has smaller indexes than $(q(1), \dots, q(m'-1))$, therefore smaller indexes than q .

□

This ends the proof of Termination Theorem (2.4).

4. APPENDIX: PDC AND SCT ARE EQUIVALENT GIVEN INJECTIVITY

The goal of this section is to compare PDC with the GTC with and without injectivity. We Claim:

1. We prove that SCT, GTC are equivalent with PDC given injectivity:

$$\forall \mathcal{R}. \mathcal{R} \text{ injective} \implies (SCT \Leftrightarrow PDC) \wedge (GTC \Leftrightarrow PDC)$$

2. Instead, in general we only have left-to-right implication

$$\forall \mathcal{R}. (SCT \Rightarrow PDC) \wedge (GTC \Rightarrow PDC)$$

3. while the opposite implication can fail:

$$\exists \mathcal{R}. \neg(\mathcal{R} \text{ injective}) \wedge \neg SCT \wedge \neg GTC \wedge PDC$$

The implication $GTC \Rightarrow PDC$ follows from the fact that for all $(S, P) \in \mathcal{R}$ there is some infinitely progressing trace $z_1, z_2, z_3, \dots$ in the path $R, R, R, R, \dots$ where $R = |(S, P)|$. In any progressing point we have $z_i P z_{i+1}$ and $z_{i+1} R z_{i+2}$. We conclude that $z_i P z_{i+1}$ and $z_i, z_{i+1} \in \text{dom}(R)$ for some z_i, z_{i+1} , that is, PDC.

In order to prove $SCT \Rightarrow PDC$, we recall that Neil Jones proved the equivalence of SCT and GTC without any assumption on $\mathcal{R}$:

$$\forall \mathcal{R}. (SCT \Leftrightarrow GTC)$$

Thus, from $GTC \Rightarrow PDC$ just proved we deduce $SCT \Rightarrow PDC$. We proved the property 2..

Now we prove the property 1.. We first prove that if $\mathcal{R}$ is injective then $PCD \Rightarrow SCT$. It will follow if $\mathcal{R}$ is injective then $PDC \Rightarrow GTC$. We just proved the opposite implications.

So it is enough to prove: $(\mathcal{R} \text{ is injective}, PCD \Rightarrow SCT)$. By Ramsey Theorem, for all binary relations R on $X = \{x_1, \dots, x_n\}$ there is $m \in \mathbf{N}$, $m \geq 1$ such that R^m is idempotent, namely such that $R^m = R^m R^m$. If we R is injective, then R^m is the identity on $\text{dom}(R^m)$, as the next Lemma says.

Proposition 4.1 (Idempotent Injective Relations and the Global Trace Condition). Assume a set $\mathcal{R}$ of bi-colored *injective* relations. Assume $(S, P) \in \mathcal{R}$, $R = |(S, P)|$ and $RR = R$. Then:

1. $R[\text{dom}(R)]$ is a bijection
2. $R[\text{dom}(R)]$ is the identity
3. If PDC holds, then some $x \in \text{dom}(R)$ is progressing to itself and all $x \in \text{dom}(R)$ are weakly progressing to themselves.

Proof. 1. We have to prove that $R[\text{dom}(R)]$ is a bijection. Let $\text{dom}(R) = \{y_1, \dots, y_m\}$. Assume $i \in [1, m]$, $y_i \in \text{dom}(R)$: from $R = RR$ we deduce that $y_i \in \text{dom}(RR)$, therefore there are $(y_i, y) \in R$ and $(y, z) \in R$. From $(y, z) \in R$ we obtain $y \in \text{dom}(R)$. Let $C_i = \{y \in \text{dom}(R) | (y_i, y) \in R\}$ and $c_i = \text{Card}(C_i)$. By $(y_i, y) \in R$ we have $c_i \geq 1$, hence $\sum_{i \in [1, m]} c_i \geq m$. By injectivity of R if $C_i \cap C_j \neq \emptyset$ then $i = j$, therefore $\{C_i | i \in [1, m]\}$ are a partition of some subset of $\text{dom}(R)$ and we have $\sum_{i \in [1, m]} c_i \leq m$. We deduce that $\sum_{i \in [1, m]} c_i = m$, and from it that $c_i = 1$ for all $i \in [1, m]$. Thus, $R[\text{dom}(R)]$ is an injection on a finite set $\text{dom}(R)$. By the characterization of bijections on finite sets we conclude that $R[\text{dom}(R)]$ is a bijection.

2. In the point above we proved that $R[\text{dom}(R)]$ is a bijection and we have to prove that *it is an identity*. Any bijection of the finite set $\text{dom}(R)$ is an union of pairwise disjoint cycles of the form $D = \{z_1, \dots, z_p\}$ for some $p \geq 1$. We have $z_i R z_j$ if and only if $j = i + 1 \pmod p$ and we have $z_i R R z_j$ if and only if $j = i + 2 \pmod p$. From $R = R R$ and $z_1 R z_2$ we deduce $z_1 R R z_2$, then by choosing $i = 1, j = 2$ that $2 = 1 + 2 \pmod p$, and eventually that $0 = 1 \pmod p$. We conclude that $p = 1$. Thus, all cycles in $R[\text{dom}(R)]$ are unary and therefore $R[\text{dom}(R)]$ is the identity.
3. (1) We prove that *for some $x \in \text{dom}(R)$, x is progressing to itself*. By PDC there are some $y, z \in \text{dom}(R)$ such that y is progressing to z . From $R[\text{dom}(R)]$ identity (the previous point) we deduce that $y = z$ for some $x \in \text{dom}(R)$. Thus, some $y \in \text{dom}(R)$ is progressing to itself.
- (2) We prove that *for all $x \in \text{dom}(R)$, x is weakly progressing to itself*. From $R[\text{dom}(R)]$ identity (the previous point) we deduce that $x R x$, either x is progressing to x or x is weakly progressing to x . In both cases x is weakly progressing to x . □

We already said that for all finite relations R there is $m \in \mathbf{N}$, $m \geq 1$ such that $R^m = R^m R^m$. By the previous Lemma, if $\mathcal{R}$ is injective, $(S, P) \in \mathcal{R}$, $R = |(S, P)|$ we deduce that some x in progressing to itself in R^m . This proves SCT, hence GTC.

The last property, namely $\exists \mathcal{R}. \neg(\mathcal{R} \text{ injective}) \wedge \neg SCT \wedge \neg GTC \wedge PDC$, was already proved with the counter-example $\mathcal{R}_2$ in S 2.3.1.

5. APPENDIX: COMPARING GLOBAL TRACE CONDITION (GTC) WITH TERMINATION OF COMPUTATIONS (TC)

In this section we prove that the Global Trace Condition is equivalent to the Termination of all possible computation flows, from now on, just TC for short.

Using Classical Logic, it is immediate that $GTC \implies TC$. Indeed, in any infinite computation flow there is some variable $x \in X$ with some infinitely progressig trace. This implies that the value of x is infinitely decreasing in the domain $\mathbf{N}$, and this cannot be.

The converse $TC \implies GTC$ is true but not immediate. Using Classical Logic, we prove it in the form $\neg GTC \implies \neg TC$. Actually we prove a stronger result: if GTC is false, there is some *constant* infinite transition $s, s, s, \dots$ for some $s = (S, P), \sigma \in \mathcal{S}$, in particular Termination of Computations (TC) fails.

$\neg GTC \implies \neg TC$. Assume $\neg GTC$. By Neil Jones Thm we deduce $\neg SCT$, namely that there is some $(S, P) \in \mathcal{R}^+$ such that for no $m \in \mathbf{N}$, $m > 0$, $x \in X$ we have x progressing to itself in $(S, P)^m$.

We will define some *constant* infinite transition $\theta = (s, s, s, \dots)$ for some state $s = (S, P), \sigma \in \mathcal{S}$.

We say that a trace $\tau : [1, t] \rightarrow X$ is irredundant if and only if all its elements are pairwise distinct, namely: $\forall i, j \in [1, t]. \tau(i) = \tau(j) \implies i = j$. An irredundant trace has at most n elements, where $n = \text{Card}(X)$, and progresses at most $n - 1$ times, because the last element of the trace does not progress, having no next element.

Let $\pi = (S, P), (S, P), \dots$ be the infinite constant path repeating (S, P) . We set $\sigma(x) =$ the maximum number progressions in an irredundant trace τ from x compatible with

π . The maximum exists because an irredundant trace has at most n elements, therefore there are finitely many irredundant traces from x . In fact, we also know that $\sigma(x) \leq n-1$.

Now we prove that $\theta = (s, s, s, \dots)$ is a constant infinite transition. We have to prove that for all $x, y \in X$ if xSy then $\sigma(x) \geq \sigma(y)$ and if xPy then $\sigma(x) > \sigma(y)$, namely $\sigma(x) \geq \sigma(y) + 1$. Assume $\sigma(x) = a$ and $\sigma(y) = b$. By definition of σ there is some irredundant trace τ from y with b progressing points in π . We have to prove $a \geq b$ if xSy and $a \geq b + 1$ if xPy .

For, we define a trace $\rho = (x)*\tau$. ρ could be not irredundant, however we can repeatedly replace each cycle included in ρ with its first element. Each time we obtain a shorter trace from x , eventually we obtain an irredundant trace ρ' . All the cycles we removed are not progressing, otherwise there would be some power of (S, P) and some $z \in X$ progressing to itself. Therefore whenever we remove a cycle from ρ we preserve the number of progressions. This number is b if xSy and is $b + 1$ if xPy . By definition of $\sigma(x) = a$ we deduce $a \geq b$ if xSy and $a \geq b + 1$ if xPy , as wished. $\square$

Remark that in fact we proved a stronger property: either we have SCT and GTC and TC, or we have $\neg$ SCT and $\neg$ GTC and $\neg$ TC. In the second case we have an infinite constant transition $\theta = (s, s, s, \dots)$ in which all traces progresses at most $n - 1$ times, with $n = \text{Card}(X)$.